

Photoabsorption and Photoionization in Gases

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Abstract—A compact demonstration paper for testing the paper compilation pathway. We summarize the distinction between photon absorption and photoionization in a dilute gas and introduce the simplest attenuation model used to connect microscopic cross sections with macroscopic interaction probabilities.

Index Terms—photoabsorption, photoionization, gas detectors, cross sections

I. INTRODUCTION

Photoabsorption denotes any process in which an incident photon transfers its energy to the medium, whereas photoionization is the subset of such interactions that leaves a free electron and a positive ion. In gaseous detectors the distinction matters because absorption controls photon transport, while ionization can seed additional charge and therefore modify the detector response.

II. MINIMAL INTERACTION MODEL

For a uniform gas with number density n , path length L , and an energy-dependent microscopic cross section $\sigma(E)$, a simple independent-interaction model gives

$$P(E, L) = 1 - \exp(-n\sigma(E)L). \quad (1)$$

Using $\sigma_{\text{abs}}(E)$ gives the total absorption probability, while $\sigma_{\text{ion}}(E)$ describes only the ionizing branch. Consequently, $\sigma_{\text{ion}}(E) \leq \sigma_{\text{abs}}(E)$ whenever non-ionizing absorption channels are available.

TABLE I
ILLUSTRATIVE QUALITATIVE BEHAVIOUR; ENTRIES ARE INTENTIONALLY SCHEMATIC.

Photon regime	σ_{abs}	σ_{ion}
Below threshold	finite or small	0
Near threshold	increasing	turn-on
Above threshold	channel dependent	finite

III. PHOTON TRANSPORT CONTEXT

In practical gaseous systems the relevant attenuation length depends strongly on photon energy, mixture composition, pressure, and the availability of ionizing and non-ionizing channels. The distinction becomes particularly useful when photons generated in an avalanche can either disappear harmlessly or create a new electron that participates in subsequent multiplication.

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IV. DISCUSSION

The simple exponential model above neglects scattering, secondary emission, excited-state kinetics, and spatial variations of the gas. Nevertheless, it provides a useful baseline for checking units, threshold behaviour, and the relation between microscopic cross sections and detector-scale probabilities. A realistic implementation can replace the schematic cross sections by tabulated or calculated data without changing the structure of the transport model.

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V. SCOPE

This document is deliberately synthetic and exists to exercise the shared paper infrastructure, metadata parsing, style switching, two-column Elsevier layout, and output routing. Its scientific content can later be replaced by a full treatment without changing the compilation workflow.

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